CLINICAL · NOSE & SINUSES

Epistaxis

Bleeding from the nose is a sign, not a diagnosis. Most bleeds are small, anterior and self-limiting; a few are posterior, brisk and dangerous. The skill lies in telling the two apart, and in climbing the ladder of treatment one rung at a time.

AT A GLANCE

- Resuscitate while you stop the bleeding.
- Pressure, vasoconstrictor, cautery, then packing.
- Then ligation or embolisation.
- Check anticoagulants and blood pressure.
- Unilateral bleeding with blockage needs a scope.
- SPA dominant in 85%; landmark the crista ethmoidalis; electrocautery beats silver nitrate.

1 CHAPTER ONE Overview

A common bleed with an uncommon tail.

1.1 Definition

Epistaxis is bleeding from inside the nose. The blood may run forward out of the nostril, backward into the nasopharynx and throat, or both. It is always a sign of something else, never a final diagnosis, and Dhingra's rule holds on every shift: an attempt should always be made to find its local or systemic cause.

1.2 Epidemiology

Most people have at least one nosebleed in their lives, and Lee estimates that around one in six of the population bleeds in any given year. Only a small fraction ever seek medical help. The distribution is bimodal: children and young adults, who bleed from the front of the septum, and adults past middle age, who make up most hospital admissions and most posterior bleeds.

1.3 Why it matters

The patients who reach the emergency department are older, often hypertensive, frequently on an anticoagulant or antiplatelet drug, and bleeding from further back in the nose, where blood runs silently down the pharynx and is easy to underestimate. In them, epistaxis is as much a resuscitation problem as a nasal one.

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2 CHAPTER TWO Applied anatomy

Two carotid systems meeting in the nose.

2.1 Arterial supply

The nasal cavity draws blood from both the internal and the external carotid arteries, and the two systems anastomose freely on the septum and the lateral wall. That double supply is why nosebleeds can be persistent, and why ligating one vessel sometimes isn't enough.

2.1.1 Internal carotid system

- **Anterior ethmoidal artery,** from the ophthalmic artery, entering the nose through the roof.
- **Posterior ethmoidal artery,** also from the ophthalmic, smaller and further back.

These supply the upper septum and lateral wall above the middle turbinate.

2.1.2 External carotid system

- **Sphenopalatine artery,** terminal branch of the maxillary artery: the main supply of the nose.
- **Greater palatine artery,** from the maxillary, reaching the septum through the incisive canal.
- **Superior labial artery,** from the facial, supplying the front of the septum and vestibule.

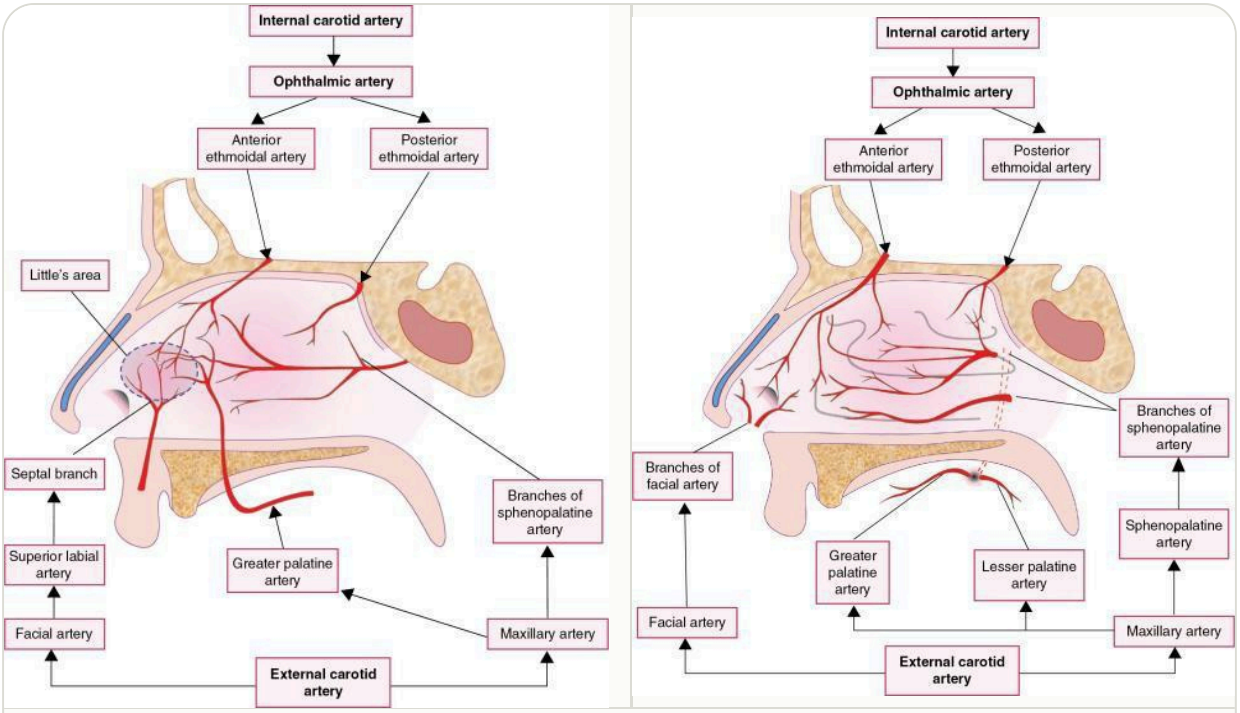


FIGURE 1 Arterial supply of the nasal septum (first) and the lateral nasal wall (second). The anterior ethmoidal, septal branch of the sphenopalatine, greater palatine and superior labial arteries converge on Little's area.

Dhingra, figs 33.1 and 33.2

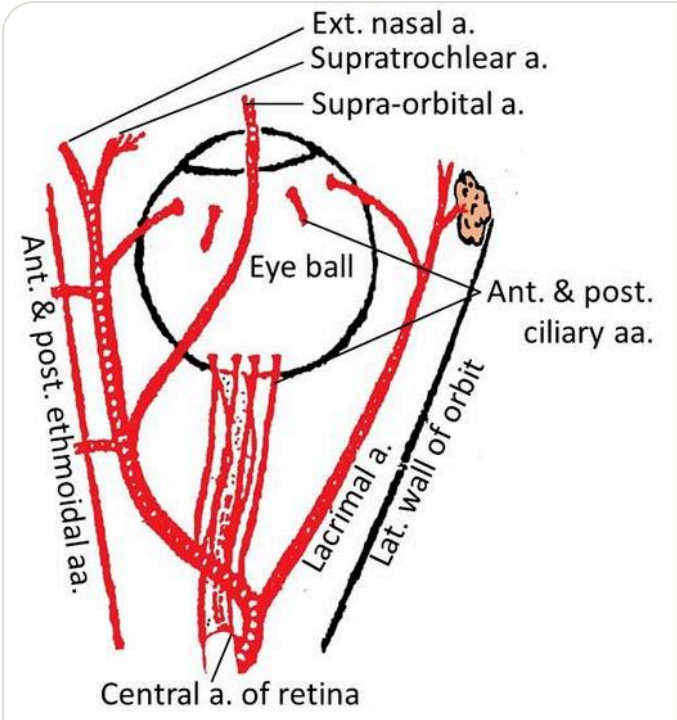


FIGURE 2 The ophthalmic artery giving off the anterior and posterior ethmoidal arteries in the orbit, on their way to the roof of the nose.

Anatomy atlas · Doss p. 100

2.2 Little's area and Kiesselbach's plexus

On the anteroinferior septum, just above the vestibule, four arteries meet to form *Kiesselbach's plexus*; the region it occupies is *Little's area*.

2.2.1 The four contributors

- Anterior ethmoidal (internal carotid)
- Septal branch of the sphenopalatine (external carotid)
- Greater palatine (external carotid)
- Septal branch of the superior labial (external carotid)

2.2.2 Why it bleeds

It sits exactly where inspired air dries and cracks the mucosa, where the mucosa is thin and closely bound to cartilage, and where a fingernail reaches. About nine in ten bleeds start here.

2.3 The posterior territory

2.3.1 Sphenopalatine artery

It enters the nose through the sphenopalatine foramen, just behind the *crista ethmoidalis* of the palatine bone, and divides into the posterior lateral nasal and posterior septal arteries. Bleeding from these branches is the classic posterior epistaxis.

2.3.2 Woodruff's plexus

A plexus on the lateral wall below the posterior end of the inferior turbinate, and the other classic posterior bleeding site in adults.

2.3.3 Ethmoidal territory

Bleeding above the level of the middle turbinate points to the ethmoidal arteries and therefore to the internal carotid system, which matters because embolisation cannot safely reach it.

WHERE THE BOOKS DIFFER

Dhingra	Lee
Woodruff's plexus is a plexus of veins below the posterior end of the inferior turbinate.	Woodruff's is the naso-nasopharyngeal plexus over the posterior centimetre of the

nasal floor, inferior meatus and middle meatus.

For the exam, give Dhingra's definition, and add that posterior arterial bleeds come from sphenopalatine branches.

2.4 Venous drainage

Veins follow the arteries. Dhingra highlights the *retrocolumellar vein*, running vertically just behind the columella across the floor of the nose, as a common source of venous bleeding in young people.

3 CHAPTER THREE Aetiology

Local, systemic, or neither.

CAUSES OF EPISTAXIS

- LOCAL **Trauma** inflammation and drying, structural, neoplastic, iatrogenic
- SYSTEMIC **Drugs** coagulation and platelet disorders, vascular disorders, cardiovascular, organ failure
- IDIOPATHIC **No cause found** despite assessment

3.1 Local causes

3.1.1 Trauma

The commonest cause overall. Digital trauma (nose picking) dominates in children; in adults add forceful blowing, blows to the face, fractures of the nasal bones, midface and skull base, and foreign bodies.

3.1.2 Inflammation and drying

Dry, heated air in winter, acute viral rhinitis and sinusitis, rhinitis sicca, atrophic rhinitis, and the crust-forming granulomatous infections all leave a fragile, cracked mucosa.

3.1.3 Structural

A septal spur or deviation creates turbulent, drying airflow over its edge; a septal perforation bleeds from its crusted rim.

3.1.4 Neoplastic

- **Benign:** juvenile nasopharyngeal angiofibroma, haemangioma, inverted papilloma.
- **Malignant:** carcinoma of the nose, sinuses and nasopharynx.

Any unilateral, recurrent bleed, especially with obstruction, must be examined endoscopically.

3.1.5 Iatrogenic and topical

Nasal and sinus surgery, nasogastric and nasotracheal tubes, nasal oxygen cannulae, topical steroid and decongestant sprays, and intranasal cocaine.

3.2 Systemic causes

The logic is simple: anything that impairs clotting, platelets or vessel walls will show up in the nose.

3.2.1 Drugs and supplements

Warfarin and the direct oral anticoagulants, aspirin, clopidogrel and other antiplatelets, NSAIDs, and, as Lee reminds us, over-the-counter vitamin E, fish oil, garlic, ginkgo and ginseng.

3.2.2 Coagulation and platelet disorders

Haemophilia, von Willebrand disease (the commonest inherited bleeding disorder), thrombocytopenia, leukaemia and aplastic anaemia.

3.2.3 Vascular disorders

Hereditary haemorrhagic telangiectasia (see
8.3
), scurvy, and the vessel fragility of old age.

3.2.4 Cardiovascular

Hypertension and atherosclerosis are associated with posterior bleeds in older adults. Whether raised pressure causes the bleed or simply keeps it going is still debated; either way, it should be measured and treated.

3.2.5 Organ failure

> Liver failure (reduced clotting factors) and renal failure (uraemic platelet dysfunction).

3.3 Idiopathic

In a large share of patients no specific cause is found despite a careful history, examination and blood tests. The label is earned only after the search, not instead of it.

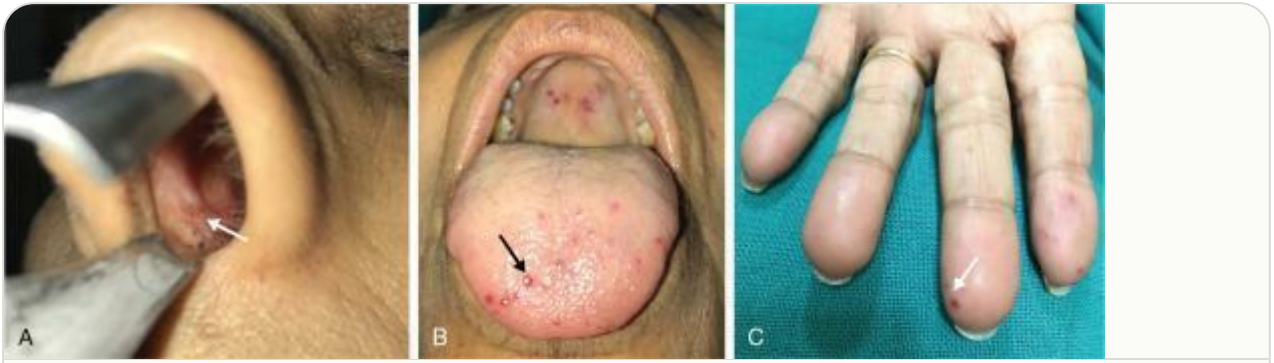


FIGURE 3 Hereditary haemorrhagic telangiectasia: lesions on the nasal septum, tongue, palate and fingertips (arrows). Always look at the lips, tongue and fingers of a recurrent bleeder.
Dhingra, fig. 33.3

RED FLAGS

- **Adolescent boy, unilateral bleeding and obstruction:** juvenile angiofibroma until proven otherwise. No biopsy in clinic.
- **Adult with unilateral bleeding, obstruction, crusting or facial pain:** endoscopy for a sinonasal tumour.
- **A child under two:** rare; think of non-accidental injury or a bleeding disorder.
- **Massive or recurrent bleeding after trauma or skull base surgery:** carotid pseudoaneurysm; needs angiography.

4 CHAPTER FOUR Classification

The distinction that drives everything that follows.

4.1 By site: anterior or posterior

With the patient sitting up, an anterior bleed runs mostly out of the nostril. A posterior bleed runs back into the throat, often from both sides, and no source is seen on anterior rhinoscopy.

TABLE

	ANTERIOR	POSTERIOR
Frequency	Common, about 90%	Less common
Site	Little's area or the front of the lateral wall	Posterosuperior cavity; often hard to localise
Age	Children and young adults	Usually over 40
Cause	Mostly trauma, dryness	Often spontaneous; hypertension, atherosclerosis, anticoagulation
Severity	Usually mild; pressure, cautery or an anterior pack	Often severe; admission, posterior pack or surgery

4.2 By severity

There is no universal scale; in practice the useful split is whether first aid is enough.

4.2.1 Minor

Stops with pressure or a single cautery, haemodynamically stable, can go home.

4.2.2 Major

Continues despite first aid, needs packing, shows haemodynamic change or a falling haemoglobin, or occurs in a frail or anticoagulated patient. These patients are admitted.

4.3 By pattern: single or recurrent

Recurrence changes the question from how to stop it to why it keeps happening. Recurrent unilateral bleeding needs endoscopy; recurrent bilateral bleeding, especially with a family history, needs HHT to be considered.

5 CHAPTER FIVE Clinical evaluation

Resuscitate first, then find the bleeding point, then ask why.

5.1 Immediate assessment

The first question is not where the blood is coming from, but how much has been lost and whether the airway is safe. A posterior bleed can fill the stomach before it fills a basin, and swallowed blood that returns as dark vomit is easily mistaken for haematemesis.

ON THE WARD: THE FIRST FIVE MINUTES

1. Sit the patient up and leaning forward. Check pulse, blood pressure and saturation; secure IV access and send a full blood count, group and save, and clotting if anticoagulated or bleeding heavily.
2. Firm pressure on the soft lower third of the nose for at least five minutes without peeking; many units ask for 10 to 15. Ice in the mouth adds a little reflex vasoconstriction.
3. Suction the clots, spray topical vasoconstrictor and anaesthetic, and look with a headlight and speculum.
4. Ask about anticoagulants, antiplatelets, previous bleeds, family history and alcohol, and which side bled first.
5. Don't stop or reverse anticoagulants for a bleed that local measures can control. Discuss with the prescribing team if it is life-threatening.

5.2 History

5.2.1 The bleed

Side of onset, duration, estimated volume, whether it runs forward or into the throat, and previous episodes and their treatment.

5.2.2 Local symptoms

Obstruction, discharge, crusting, facial pain or numbness, and trauma.

5.2.3 Systemic background

Medications including supplements, hypertension, liver or renal disease, easy bruising, and a family history of bleeding or nosebleeds.

5.3 Examination

5.3.1 General

Pallor, pulse and blood pressure, signs of liver disease, bruising, and telangiectasias on the lips, tongue and fingertips.

5.3.2 Anterior rhinoscopy

After clot removal and vasoconstriction: inspect Little's area first, then the front of the lateral wall. The oropharynx is checked for blood running down from behind.

5.3.3 Nasal endoscopy

For recurrent bleeds, for bleeding not localised on rhinoscopy, and for any unilateral bleed with suspicious symptoms. It finds points hidden lateral to the turbinates or high in the nose and allows directed cautery instead of a blind pack.

5.4 Investigations

5.4.1 Blood tests

Not needed for a minor anterior bleed in a healthy patient. For significant bleeds: full blood count, group and save or crossmatch, clotting with INR in warfarinised patients, and liver and renal function where relevant.

5.4.2 Imaging

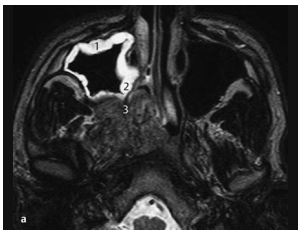
CT or MRI when a tumour is suspected; for juvenile angiofibroma, contrast imaging replaces biopsy.

5.4.3 Angiography

Diagnostic and therapeutic: for suspected pseudoaneurysm, vascular tumours, and as the first step of embolisation.

IMAGING IN THE APP 1 case

Juvenile angiofibroma, the tumour to think of in a teenage boy with one-sided bleeding.



Angiofibroma, Juvenile

Tumours · 12 images

6 CHAPTER SIX Local management

The ladder of control.

6.1 The treatment ladder

1. **First aid and topical vasoconstriction**
   Pressure on the soft nose, leaning forward; pledgets of vasoconstrictor and local anaesthetic.
2. **Cautery of a visible point**
   Silver nitrate or bipolar, to the bleeding point only.
3. **Anterior packing**
   Ribbon gauze, expanding sponge, or resorbable material.
4. **Posterior packing**
   Postnasal pack, Foley or dual balloon; admit and monitor.
5. **Endoscopic cautery**
   Directed at the source under local or general anaesthesia.

6 **Ligation or embolisation**
For bleeding that persists despite packing.

6.2 First aid

Trotter's method: the patient sits up, leaning slightly forward over a basin, breathing through the mouth, while the soft part of the nose is pinched firmly against the septum. Leaning forward stops blood being swallowed and lets you see how much is lost. Cold compresses add reflex vasoconstriction.

6.3 Cautery

Cautery works when you can see what you are treating. Anaesthetise first, and treat only the bleeding point.

6.3.1 Chemical

Roll a silver nitrate stick over a small ring around the vessel, edge first and then inwards, before touching the vessel itself. Best for small, visible anterior points in a field that is not actively pouring.

6.3.2 Electrocautery

Bipolar diathermy, usually under endoscopic view, is more effective for brisker or deeper points and for posterior sites.

PEARL

Never cauterise both sides of the septum at the same sitting: the mucoperichondrium on both sides may necrose and leave a perforation.

6.4 Anterior packing

6.4.1 Ribbon gauze

About a metre of ribbon gauze, 2.5 cm wide in adults and 12 mm in children, soaked in liquid paraffin or bismuth iodoform paraffin paste and laid in layers so that it presses evenly on the whole cavity. Usually removed after 24 hours; antibiotics are given if it is left longer.

6.4.2 Expanding sponges and balloons

Compressed sponges that swell when wetted, and inflatable balloon packs: quicker to insert, more comfortable, and now the standard in most emergency departments.

6.4.3 Resorbable packs

Preferred when the patient is anticoagulated or has a bleeding disorder, because removing a non-resorbable pack can restart the bleed.

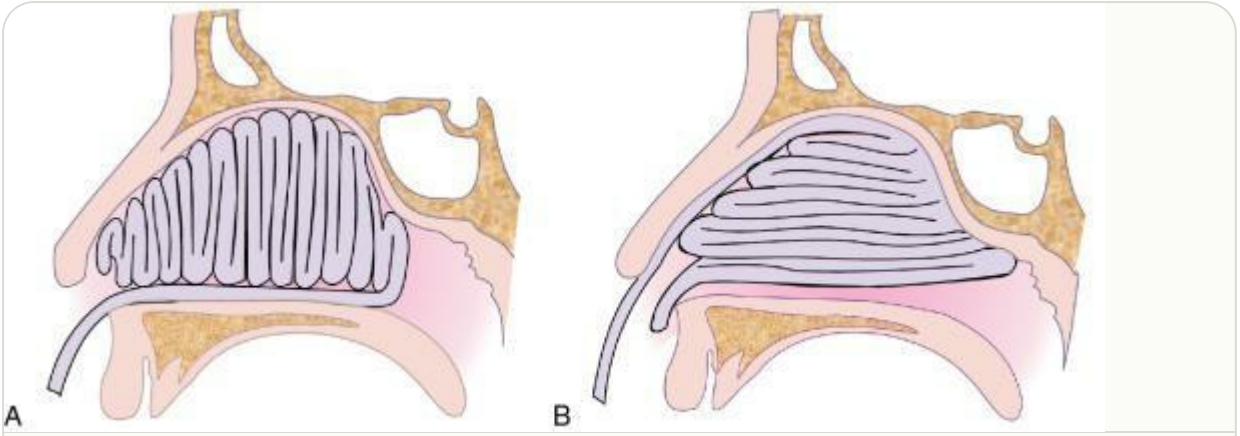


FIGURE 4 Anterior nasal packing. (A) Vertical layers from the back forwards. (B) Horizontal layers from the floor upwards.
Dhingra, fig. 33.4

6.5 Posterior packing

When blood runs back into the pharynx and an anterior pack fails, the choana must be blocked.

6.5.1 Postnasal gauze pack

A rolled gauze cone with three silk ties is drawn into the nasopharynx on a catheter passed through the nose and out of the mouth, then pushed into place with a finger. An anterior pack goes in front, and two ties are secured over a dental roll at the columella. The third tie is left in the oropharynx, taped to the cheek, to pull the pack out later.

6.5.2 Foley catheter

A 12 to 14 F Foley is passed until the balloon lies in the nasopharynx, inflated with saline (not air), drawn forward against the choana and backed by an anterior pack. Protect the ala and columella from pressure.

6.5.3 Dual-balloon devices

Commercial catheters with a posterior and an anterior balloon, and a central tube that preserves a nasal airway.

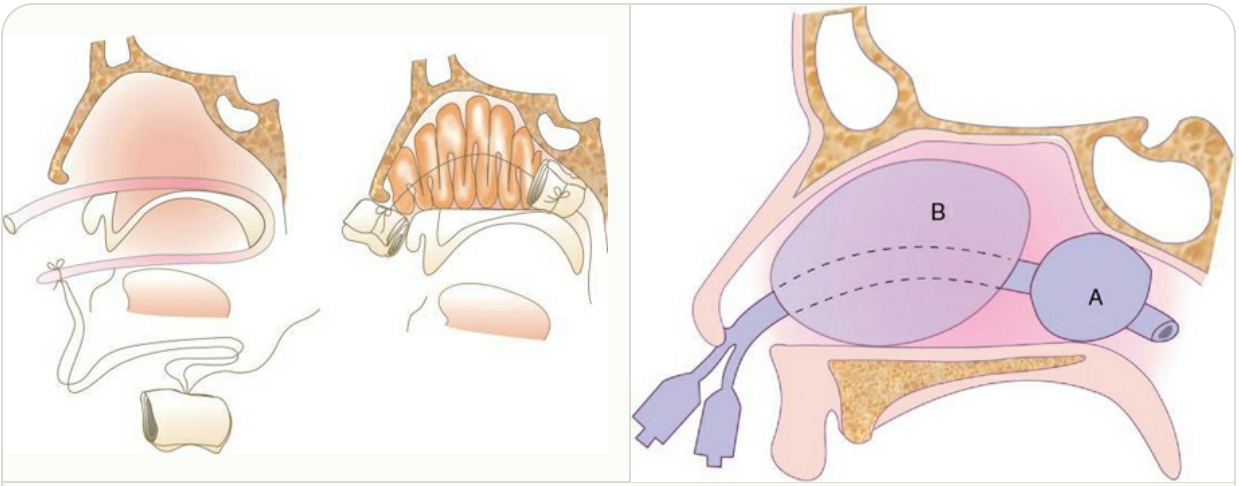


FIGURE 5 First, inserting a postnasal pack on a catheter. Second, a dual-balloon catheter with the posterior balloon at about 10 mL and the anterior at about 30 mL.
Dhingra, figs 33.5 and 33.6

6.5.4 Care of the packed patient

- **Admit and monitor:** posterior packs fail in about a third.
- **Oxygen:** bilateral packs can cause hypoxia through the nasopulmonary reflex, especially in the elderly.
- **Watch for:** alar and columellar necrosis, sinusitis, otitis media, aspiration and toxic shock syndrome.
- **Antibiotics** are traditionally given while a non-resorbable pack is in, though the evidence for routine use is weak.
- **Explain the pack:** what type it is, when it comes out, and what should bring them back.

6.6 Medical adjuncts

6.6.1 Blood pressure

Treat pain and anxiety first; control persistent hypertension, but avoid dropping the pressure sharply.

6.6.2 Tranexamic acid

Topical tranexamic acid did not reduce the need for packing in the NoPAC trial; it should not delay standard treatment.

GUIDELINE UPDATE

The **AAO-HNS Nosebleed guideline (2020)**: compression for at least five minutes; packing if the site cannot be seen; resorbable packs on anticoagulants or antiplatelets; endoscopy for recurrent or difficult bleeds; cautery confined to the bleeding site; ligation or embolisation when these fail; and first-line local treatment before transfusion, reversal or stopping anticoagulants unless the bleed is life-threatening.

7 CHAPTER SEVEN Surgery and intervention

Taking out the supply: the vessel follows the site.

7.1 Choosing the vessel

PERSISTENT BLEEDING

- SOURCE SEEN **Endoscopic cautery** under local or general anaesthesia
- POSTERIOR, BELOW THE MIDDLE TURBINATE **Sphenopalatine ligation** or embolisation
- ABOVE THE MIDDLE TURBINATE **Ethmoidal ligation** not embolisation
- SEPTAL DEVIATION **Septal surgery** to reach the bleed

7.2 Endoscopic sphenopalatine artery ligation

7.2.1 Indications

Posterior bleeding that persists or recurs despite adequate packing. TESPAL is now the operation of choice and has replaced the older open procedures.

7.2.2 Technique

Under general anaesthesia, a mucosal incision on the lateral wall in front of the crista ethmoidalis; a mucoperiosteal flap is raised back to the crista, behind which the artery and its branches emerge from the foramen. They are clipped or coagulated, and the flap is replaced.

7.2.3 Outcomes and complications

Lee quotes control in 95 to 98%, usually with discharge the next day. Complications are minor: crusting, adhesions and numbness of the palate or nose.

7.3 Ethmoidal artery ligation

7.3.1 Approach

An external (Lynch) incision at the medial canthus, raising the periosteum of the medial orbital wall, usually for high bleeds after trauma.

7.3.2 The 24-12-6 rule

The anterior ethmoidal foramen lies about 24 mm behind the anterior lacrimal crest, the posterior foramen about 12 mm further back, and the optic canal about 6 mm beyond that. The posterior artery is approached with great care, or left alone.

7.4 Embolisation

7.4.1 Method

The internal maxillary artery is catheterised by the femoral route and blocked with particles, gelfoam or coils, often bilaterally because of cross-flow.

7.4.2 Who it suits

Patients unfit for anaesthesia, vascular tumours and traumatic pseudoaneurysms. It cannot treat the ethmoidal arteries, which arise from the internal carotid system.

7.4.3 Outcomes and risks

Success around 88%. Complications include stroke, blindness, facial pain and skin necrosis.

7.5 Septal surgery

Raising a mucoperichondrial flap divides the vessels feeding a recurrent septal bleed; septoplasty also straightens a deviation that hides a bleeding point or

prevents adequate packing.

7.6 Historical operations

7.6.1 Maxillary artery ligation

Through a Caldwell-Luc approach and the posterior antral wall. Failure in 10 to 15%, with infraorbital numbness, oroantral fistula and dental injury.

7.6.2 External carotid ligation

Above the origin of the superior thyroid artery. Too proximal to be reliable, given the collateral flow; of examination interest now.

PEARLS

- Bleeding above the middle turbinate is ethmoidal: embolisation won't help, think ligation.
- The crista ethmoidalis is the landmark for the sphenopalatine foramen: the artery lies just behind it.
- If endoscopic ligation fails, look for an ethmoidal source or a missed accessory branch before repacking.

OPERATIVE GUIDES IN THEATRE 4 operations

Anterior and Posterior
Anterior and posterior ethmoid artery ligation
Sphenopalatine Artery Ligation for Epistaxis
Sphenopalatine artery ligation
Surgical Management of Hereditary Hemorrhagic Telangiectasia
HHT
Juvenile Angiofibroma
Angiofibroma

8 CHAPTER EIGHT Special situations

Children, the anticoagulated, HHT, and the bleed after trauma.

8.1 Children

8.1.1 Typical picture

Bleeding from Little's area, from crusting and picking, that stops on its own.

8.1.2 Treatment

Antiseptic cream such as chlorhexidine with neomycin to the vestibule twice daily for four weeks settles most cases; silver nitrate for a visible vessel that keeps bleeding. Avoid creams containing arachis oil in peanut allergy.

8.1.3 When to investigate

Recurrent bleeding with easy bruising or a family history needs a clotting screen including von Willebrand studies. Under two years of age, think of a bleeding disorder or non-accidental injury.

8.2 The anticoagulated patient

Patients on warfarin, a direct oral anticoagulant or dual antiplatelets bleed longer and re-bleed more often, but stopping their drugs risks stroke or stent thrombosis. Control the bleed locally first with resorbable packing and check the INR in warfarinised patients. Reversal and withholding are for life-threatening bleeds, decided with the prescribing team.

8.3 Hereditary haemorrhagic telangiectasia

Recurrent bleeding from multiple telangiectasias, usually from childhood, with a family history. Repeated packing and aggressive cautery make things worse; laser, bevacizumab, septodermoplasty and Young's procedure are the tools. See the HHT topic for the full account.

8.4 Trauma and pseudoaneurysm

Delayed, massive or recurrent bleeding days to weeks after head injury or skull base surgery suggests a pseudoaneurysm of the internal carotid artery. It is a neurovascular emergency: pack, resuscitate and arrange urgent angiography.

9 CHAPTER NINE Complications and outcome

Of the bleed and of its treatment.

9.1 Of the bleed

Hypovolaemic shock, anaemia, aspiration, and myocardial ischaemia in patients with coronary disease.

9.2 Of treatment

- **Cautery:** septal perforation.
- **Packing:** pressure necrosis, sinusitis, otitis media, hypoxia, toxic shock syndrome.
- **Surgery and embolisation:** see 7.2 to 7.4 .

9.3 Prognosis

Most bleeds settle with simple measures. Recurrence is common in the elderly, the anticoagulated and those with HHT, which is why the cause matters as much as the cure.

PRACTICAL POINTS (OXFORD TEXTBOOK)

About 95% of bleeds arise from Little's area, but the **sphenopalatine artery** is the dominant supply to the nose in about 85% of people (the rest are anterior ethmoid-dominant), which is why endoscopic sphenopalatine ligation works so well for posterior bleeds. At surgery the foramen lies just behind and below the **crista ethmoidalis**, a bony ridge on the lateral wall near the posterior end of the middle turbinate, and the artery often divides into several branches before emerging, all of which must be controlled. Risks to explain include rebleeding, crusting, palatal numbness, adhesions, epiphora and, very rarely, visual loss.

Where available, **electrocautery** is preferred to silver nitrate because it fails less often and reduces packing, admission and pain; after silver nitrate, irrigate with saline to

prevent skin burns. Ice on the nasal bridge has no proven benefit. A patient whose anterior bleed is stopped by cautery can go home with antiseptic or antibiotic cream for two weeks; after a significant posterior bleed, observe for 6 to 12 hours; anyone with a pack is admitted. Many specialists give antibiotics when a pack stays more than 48 hours. For warfarin with major bleeding, give intravenous vitamin K and prothrombin complex concentrate (fresh frozen plasma only if that is unavailable); for minor bleeding with a high INR, withhold warfarin and give oral vitamin K. Ask about "natural blood thinners" such as garlic, ginger, ginkgo, turmeric and vitamin E.

10 CHAPTER TEN For the examination

What the examiner wants to hear.

10.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Little's area:** the four arteries, their parent vessels, and why it bleeds.
- **Trotter's method:** position, mouth breathing, and pinching the soft nose.
- **Anterior versus posterior:** the five-row comparison.
- **Postnasal pack:** technique, the third tie, and why the patient is admitted.
- **Which artery for which site,** and why ethmoidal bleeds can't be embolised.
- **Teenage boy, unilateral recurrent bleed:** angiofibroma, and no biopsy in clinic.

10.2 Common pitfalls

COMMON PITFALLS

- Underestimating blood loss in an elderly patient who is swallowing blood.
- Packing blindly before suctioning and looking, when a cauterisable point was visible.
- Cauterising both sides of the septum at once.
- Stopping anticoagulation reflexively for a bleed a pack would have controlled.

- Repeated packing in recurrent bilateral bleeds that are really HHT.

Written for this app from Oxford Textbook of Otolaryngology ch. 27, 77, Oxford Specialist Handbook ch. 20, Oxford Handbook of ENT ch. 9, 25, Dhingra ch. 33 and K. J. Lee ch. 31. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

NOSE & NASAL CAVITY · TUMOURS

Angiofibroma, Juvenile

IMAGING FINDINGS

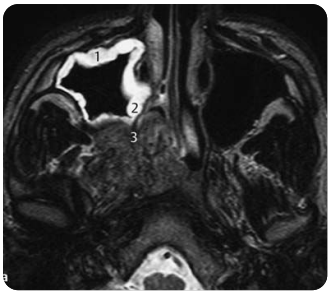
- Enhancing mass at the sphenopalatine foramen in an adolescent male
- Widened pterygopalatine fossa; bowing of the posterior maxillary wall
- Flow voids; extension to the sphenoid, infratemporal fossa and skull base

TOP DIFFERENTIALS

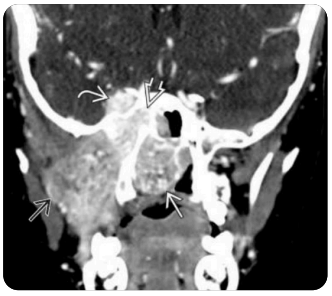
- Hypervascular polyp
- Rhabdomyosarcoma
- Antrochoanal polyp

CHECKLIST

- No biopsy: angiography and embolisation before surgery
- Vidian canal and cavernous sinus spread



A young man with obstruction and bloodtinged nasal discharge from the right side of the nose.
Dunnebier, Imaging for…



Coronal CECT in a 13-year-old shows typical growth pattern of a large, hypervascular angiofibroma occluding the nasopharynx, extending into righ…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 740.